

Chapter 7

Event Selection

This chapter defines the signal regions (SRs) targeting the processes shown in Figure 3.4. The analysis variables and preselection are introduced first, followed by the optimisation of the one- b -jet regions, the definition of the zero- b -jet regions, and a study of $\tau\tau$ signal contamination in the SRs.

7.1 Preselection and variables definition

7.1.1 Analysis variables

The transverse mass m_T of the τ and $\vec{p}_T^{\text{miss}}$ system is defined as

$$m_T = \sqrt{2p_T^\tau E_T^{\text{miss}} [1 - \cos \Delta\phi(\tau, E_T^{\text{miss}})]}. \quad (7.1)$$

Here $\Delta\phi(\tau, E_T^{\text{miss}})$ is the separation in ϕ between the τ momentum and $\vec{p}_T^{\text{miss}}$. The variable m_T is used to probe the high-energy tail of the $\tau\nu$ final state, which is particularly sensitive to non-resonant leptoquark contributions [1].

The invariant masses $m_{b\tau}$ and $m_{j\tau}$ are defined as:

$$m_{b\tau}^2 = (p_b + p_\tau)^2, \quad m_{j\tau}^2 = (p_{j_1} + p_\tau)^2, \quad (7.2)$$

where p_b , p_{j_1} and p_τ are four-momenta, and j_1 is the jet with the largest p_T . The zero- b -jet regions contain no b -tagged jet, so $m_{j\tau}$ is used in place of $m_{b\tau}$.

7.1.2 Multijet suppression and Preselection

Although the high E_T^{miss} threshold rejects most multijet events, some can still enter the SRs and CRs due to the large multijet cross section. These events typically contain a misidentified τ and fake E_T^{miss} from poor jet resolution. Therefore, further suppression is necessary.

Specifically, fake E_T^{miss} is produced when an energetic jet is mismeasured. In this case, $\vec{p}_T^{\text{miss}}$ points approximately towards that jet and it tends to be small compared with the p_T of the mismeasured jet. On the other hand, in signal events, $\vec{p}_T^{\text{miss}}$ from neutrinos has no such tendency. Figure 7.1 illustrates this difference. The minimum jet- $\vec{p}_T^{\text{miss}}$ separation in ϕ and the ratio $\frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min \Delta\phi)}}$ are therefore used as discriminating variables.

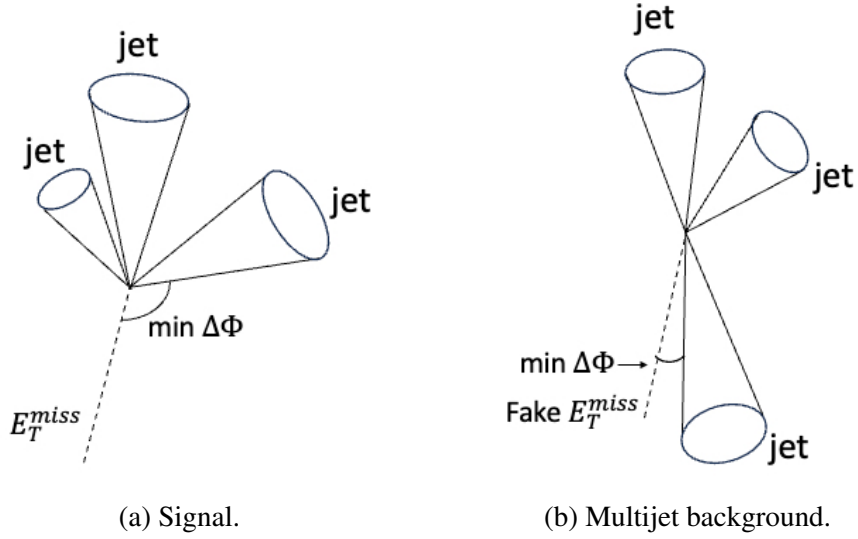


Fig. 7.1 Schematic diagrams of (a) E_T^{miss} from neutrinos in signal events and (b) fake E_T^{miss} from a mismeasured energetic jet in multijet events.

Here $p_T^{\text{jet}(\min \Delta\phi)}$ is the p_T of the jet with the smallest $\Delta\phi(E_T^{\text{miss}}, p_T^{\text{jet}})$.

Events are retained if

$$\min \Delta\phi(E_T^{\text{miss}}, p_T^{\text{jet}}) > 0.4 \quad \text{or} \quad \frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min \Delta\phi)}} > 6. \quad (7.3)$$

The study used to determine the threshold above is presented in detail in Appendix B.

In addition to the multijet suppression requirement, the following preselection requirements are applied before the SR optimisation:

- The lowest-threshold unscaled E_T^{miss} trigger to select events with large E_T^{miss} .
- $E_T^{\text{miss}} > 200$ GeV for full trigger efficiency.

- Exactly one signal τ .
- A signal electron/muon veto.
- Exactly one b -tagged jet for the one- b -jet SR (SR1b).
- At least one untagged jet and no b -tagged jets for the b -jet-veto SR (SR0b).

7.2 SR1b optimisation and definition

Event selection in SR1b is optimised using a grid scan of thresholds on variables that help separate signal from background. SR1b-Res and SR1b-NonRes are optimised separately.

7.2.1 Scan variables and criterion

p_T^τ and E_T^{miss} are considered first, since energetic τ leptons and neutrinos are expected in the signal. The b -jet p_T is also used. An energetic b -jet from the leptoquark decay is expected in the resonant process, whereas the b -jet is typically softer in the non-resonant process. m_T is used to distinguish non-resonant leptoquark exchange from the SM background. A tighter m_T threshold is therefore expected in SR1b-NonRes than in SR1b-Res. The invariant mass $m_{b\tau}$ is used to probe resonant $U_1 \rightarrow b\tau$ decays. $\Delta\phi(\tau, E_T^{\text{miss}})$ is used to select the back-to-back τ and $\vec{p}_T^{\text{miss}}$ configuration of the non-resonant signal, which helps suppress the top-quark background. On the other hand, in SR1b-Res, small $\Delta\phi(\tau, E_T^{\text{miss}})$ are favoured due to the high p_T^b requirement. The jet multiplicity is used to suppress $t\bar{t}$ events since they tend to have more jets than the signal events. The thresholds used for grid scan are summarised in Table 7.1.

Table 7.1 Variable scan range for the SR1b optimisation.

Variable	SR1b-Res	SR1b-NonRes
$\Delta\phi(\tau, E_T^{\text{miss}})$	$< \{0.3, 1.2, 2.4, 2.6, \pi\}$	$> \{0.3, 1.2, 2.4, 2.6\}$
$m_{b\tau}$ [GeV]	$> \{0, 800\}$	N/A
E_T^{miss} [GeV]	$> \{200, 250, 300, 350, 400, 500, 600\}$	
p_T^τ [GeV]	$> \{200, 250, 300, 350, 400, 500, 600\}$	
m_T [GeV]	$> \{100, 200, 300, 400, 500, 600, 700, 800, 900, 1000\}$	
$N_j - N_b$		$\leq \{1, 2, 3\}$

Note that only the $m_{b\tau} > 0$ and 800 GeV are tested here. Since the signal mass resolution is only about 50% and very few background events are remained in the SRs, little improvement in sensitivity is expected from a finer mass threshold scan.

Different signal benchmarks are selected for each signal region. The optimisation for SR1b-Res uses $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$, where on-shell production is dominant. For SR1b-NonRes, the optimisation uses $(M_U, g_U, \beta_L^{23}) = (2.5 \text{ TeV}, 2.5, 1.0)$, where t -channel exchange is dominant. Interference is not included in the scan because its contribution is around 5–10% in the parameter range probed by the SR1b regions, as discussed in Appendix D. The selected cut combination is therefore not expected to change significantly when interference is included.

The expected sensitivity of each cut combination is evaluated using the significance Z , calculated as follows [2]:

$$Z = \sqrt{2 \left[n \ln \left(\frac{n(b + \sigma_b^2)}{b^2 + n\sigma_b^2} \right) - \frac{b^2}{\sigma_b^2} \ln \left(\frac{b^2 + n\sigma_b^2}{b(b + \sigma_b^2)} \right) \right]}, \quad (7.4)$$

where s and b are the expected signal and background event yields, $n = s + b$, and the background uncertainty is assumed to be $\sigma_b = 0.3b$.

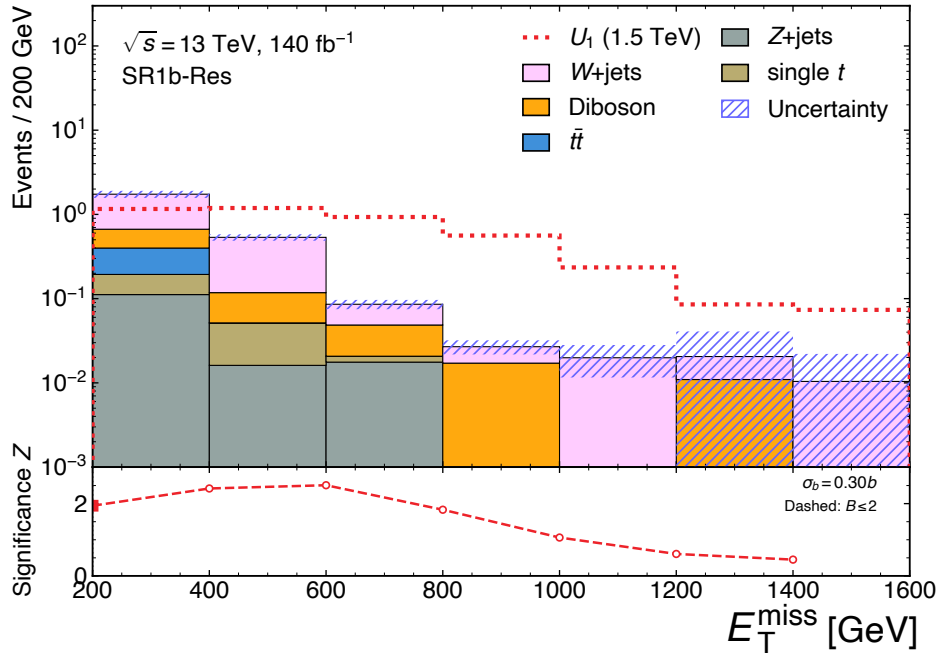
To obtain reliable results, the following two constraints are applied during the scan:

- Cut combinations with $b \leq 2$ are discarded.
- If a particular SM background (e.g. $W \rightarrow \tau\nu$) has a negative weighted yield, its contribution is set to zero.

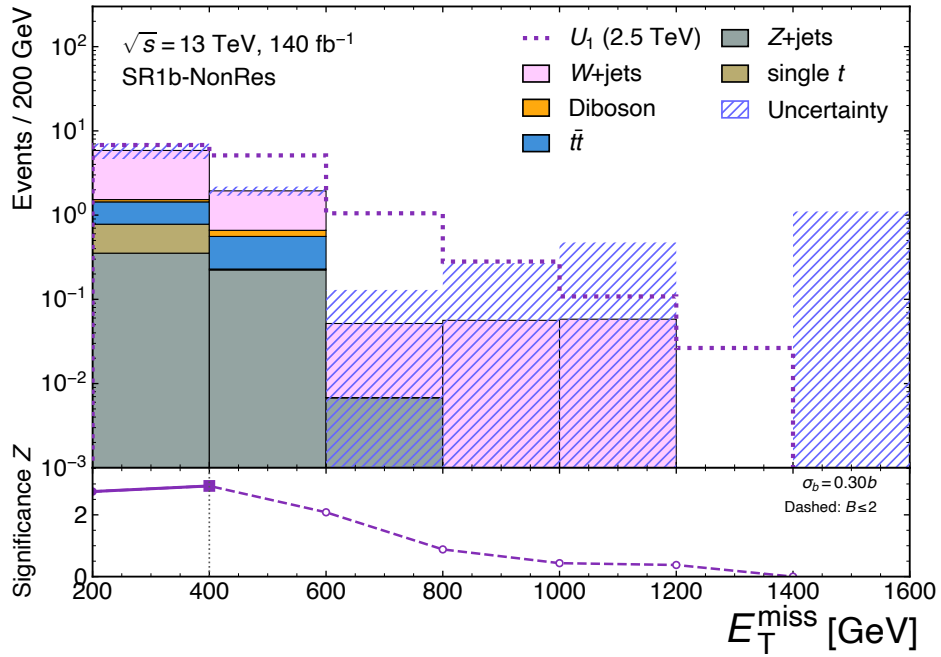
The first constraint is used to avoid zero background estimates caused by limited MC statistics. It also ensures reliable estimates of the minor backgrounds. As a result, the MC statistical uncertainties are limited to 31% in the resonant SR and 40% in the non-resonant SR.

An example of the significance scan is shown in Figure 7.2, along with $N - 1$ E_T^{miss} distributions.¹ For each bin in the lower significance panels, Z is calculated using the signal and background yields summed above the corresponding threshold. Points with $b \leq 2$ are excluded from the cut choice. The largest Z is therefore obtained at 200 GeV for SR1b-Res and 400 GeV for SR1b-NonRes. A similar procedure is applied to the other variables.

¹For a selection with N requirements, an $N - 1$ distribution is obtained by removing the requirement on the plotted variable while keeping the other $N - 1$ requirements. Here, the E_T^{miss} requirement is removed.



(a) SR1b-Res.



(b) SR1b-NonRes.

Fig. 7.2 $N - 1$ E_T^{miss} distributions and the corresponding significance Z . Z is calculated from the signal and background yields above each threshold, with $\sigma_b = 0.30b$. Dashed segments of the Z curves indicate regions where $b \leq 2$.

7.2.2 SR1b-Res definition

The definition and cutflow for SR1b-Res is summarised in Table 7.2, using the resonant optimisation benchmark $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$. A single event count with $m_{b\tau} > 800 \text{ GeV}$ is used because the simulated background statistics at high $m_{b\tau}$ are insufficient for a fit to the mass distribution.

Table 7.2 Signal and background cutflow for SR1b-Res. The signal is evaluated at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$. The uncertainties are from MC statistics only.

Requirement	Signal	$t\bar{t}$	Single top	W+jets	Z+jets	Diboson	Total background
Baseline	35.2 ± 0.4	1043.0 ± 6.9	283.8 ± 3.4	2156.9 ± 16.1	369.2 ± 7.7	173.9 ± 2.7	4027.3 ± 19.7
$p_T^\tau > 200 \text{ GeV}$	19.9 ± 0.3	32.3 ± 1.3	11.8 ± 0.5	108.9 ± 9.4	30.4 ± 4.7	9.7 ± 0.6	193.1 ± 10.6
$E_T^{\text{miss}} > 200 \text{ GeV}$	19.9 ± 0.3	32.3 ± 1.3	11.8 ± 0.5	108.9 ± 9.4	30.4 ± 4.7	9.7 ± 0.6	193.1 ± 10.6
$m_T > 200 \text{ GeV}$	19.4 ± 0.3	25.1 ± 1.1	9.2 ± 0.4	79.2 ± 9.4	24.2 ± 4.7	6.9 ± 0.5	143.6 ± 10.6
$m_{b\tau} > 800 \text{ GeV}$	4.9 ± 0.1	0.4 ± 0.2	0.4 ± 0.1	2.8 ± 0.4	1.1 ± 0.2	0.5 ± 0.1	5.2 ± 0.5
$N_j - N_b \leq 3$	4.7 ± 0.1	0.3 ± 0.1	0.2 ± 0.1	2.5 ± 0.4	0.9 ± 0.2	0.4 ± 0.1	4.3 ± 0.5
$p_T^b > 250 \text{ GeV}$	4.2 ± 0.1	0.2 ± 0.1	0.1 ± 0.1	1.5 ± 0.2	0.2 ± 0.1	0.4 ± 0.1	2.4 ± 0.3

As a result, the expected signal and background yields are 4.2 and 2.4 events, respectively, giving $Z = 1.95$.

The background is dominated by $W(\rightarrow \tau\nu)$ +jets, which contributes approximately 64%. Approximately 16% is contributed by diboson production. The top-quark contribution consists of $t\bar{t}$ (8%) and single-top production (5%), with the latter mainly from the Wt process. These events are mainly from $t\bar{t} \rightarrow b\tau\nu\bar{b}\tau\nu$ or $Wt \rightarrow b\tau\nu\tau\nu$ when only one τ is selected and the other is misidentified as a jet. A smaller contribution is provided by $Z(\rightarrow \tau\tau)$ +jets, while the $Z(\rightarrow \nu\nu)$ +jets contribution is negligibly small.

The $N - 1$ distributions in SR1b-Res with relaxed kinematic requirements are shown in Figure 7.3. To reduce statistical fluctuations, the thresholds on the correlated variables m_T and p_T^τ are both lowered to 100 GeV.

For the 1.5 TeV benchmark, the $m_{b\tau}$ peak is shifted to about 900 GeV because the neutrino from the τ decay is not included in the reconstructed mass. The mass distribution is also broadened by the missing neutrino momentum and poor jet reconstruction resolution.

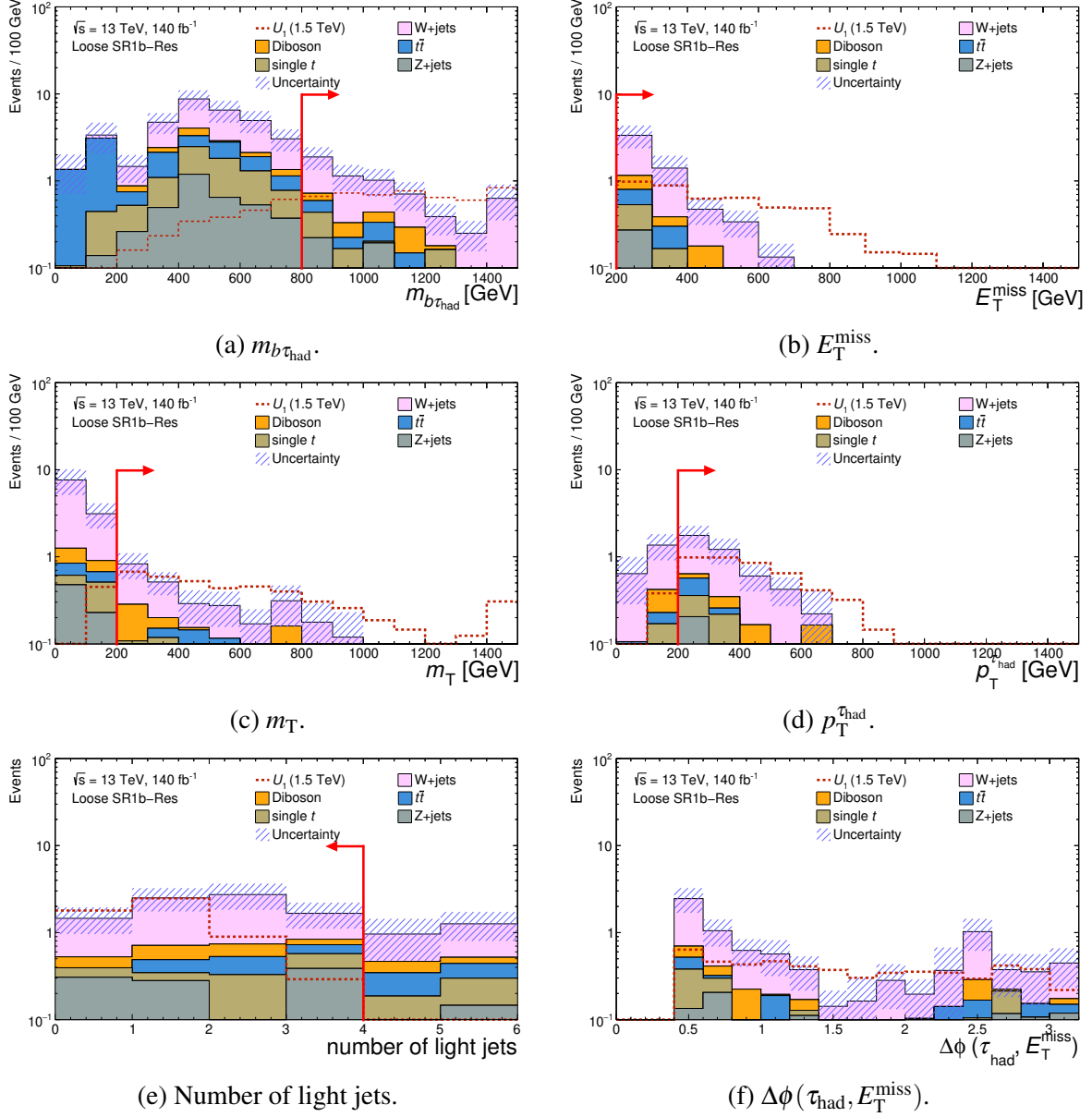


Fig. 7.3 $N-1$ distributions in the loose SR1b-Res region. The m_T and p_T^{τ} thresholds are reduced to 100 GeV. The red dashed line represents the signal at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$. The hatched band represents the background uncertainty, including systematic contributions. The red arrows mark the final selection thresholds and point towards the retained ranges.

7.2.3 SR1b-NonRes definition

SR1b-NonRes is defined by requiring $50 < p_T^b < 250$ GeV, $p_T^\tau > 200$ GeV, $E_T^{\text{miss}} > 400$ GeV, $m_T > 600$ GeV and $\Delta\phi(\tau, E_T^{\text{miss}}) > 1.2$, with at most one jet that is not b -tagged. No $m_{b\tau}$ requirement is applied because the b -jet and τ do not reconstruct an on-shell leptoquark in the target topology. This region also contributes a single event count to the fit.

The background is dominated by W +jets, principally $W \rightarrow \tau\nu$, which contributes approximately 69%. The next-largest contribution is from $t\bar{t}$ (16%), followed by $Z \rightarrow \tau\tau$ +jets (8%) and diboson production (5%). The $Z \rightarrow \tau\tau$ contribution requires only one selected τ , with neutrinos from the decays producing E_T^{miss} . In both SR1b regions, $Z \rightarrow \nu\nu$ +jets is the main source of events with a jet misidentified as a τ , because the invisible Z decay supplies the missing momentum. Multijet production is negligible after the cleaning and full SR requirements.

For the optimisation benchmark $(M_U, g_U, \beta_L^{23}) = (2.5 \text{ TeV}, 2.5, 1.0)$, the expected signal and background yields are 6.583 and 2.097 events, respectively, giving $Z = 2.94$.

Table 7.3 Signal (S) and background (B) cutflow in the SR1b-NonRes optimisation for $(M_U, g_U, \beta_L^{23}) = (2.5 \text{ TeV}, 2.5, 1.0)$.

Cumulative requirement	S	B
Baseline	111.0	4027.3
$p_T^\tau > 200$ GeV	70.5	193.1
$E_T^{\text{miss}} > 400$ GeV	26.4	15.1
$m_T > 600$ GeV	24.0	6.4
$\Delta\phi(\tau, E_T^{\text{miss}}) > 1.2$	23.8	6.4
$N_j - N_b \leq 1$	17.2	2.8
$50 < p_T^b < 250$ GeV	6.6	2.1

The yields and significances for both regions are summarised in Table 7.4. The yields agree with the rounded endpoints of the optimisation cutflows to within 2%. The cutflows describe the optimisation study, while the final selections are chosen with the statistical uncertainty in the background prediction kept below 30%.

Table 7.4 Signal and background yields for the two optimisation benchmarks in the final SR1b selections. The approximate significance is evaluated with $\sigma_b = 0.30b$.

SR	(M_U, g_U, β_L^{23})	S	B	Z
SR1b-Res	(1.5 TeV, 1.5, 0.6)	4.262	2.439	1.95
SR1b-NonRes	(2.5 TeV, 2.5, 1.0)	6.583	2.097	2.94

116 The loose $N - 1$ distributions in Figure 7.4 illustrate the high- m_T and high- E_T^{miss} tails of the
117 non-resonant signal. The m_T , E_T^{miss} and p_T^τ thresholds are each lowered by 100 GeV before the
118 plotted requirement is removed. The trigger-plateau requirement $E_T^{\text{miss}} > 200$ GeV is retained.

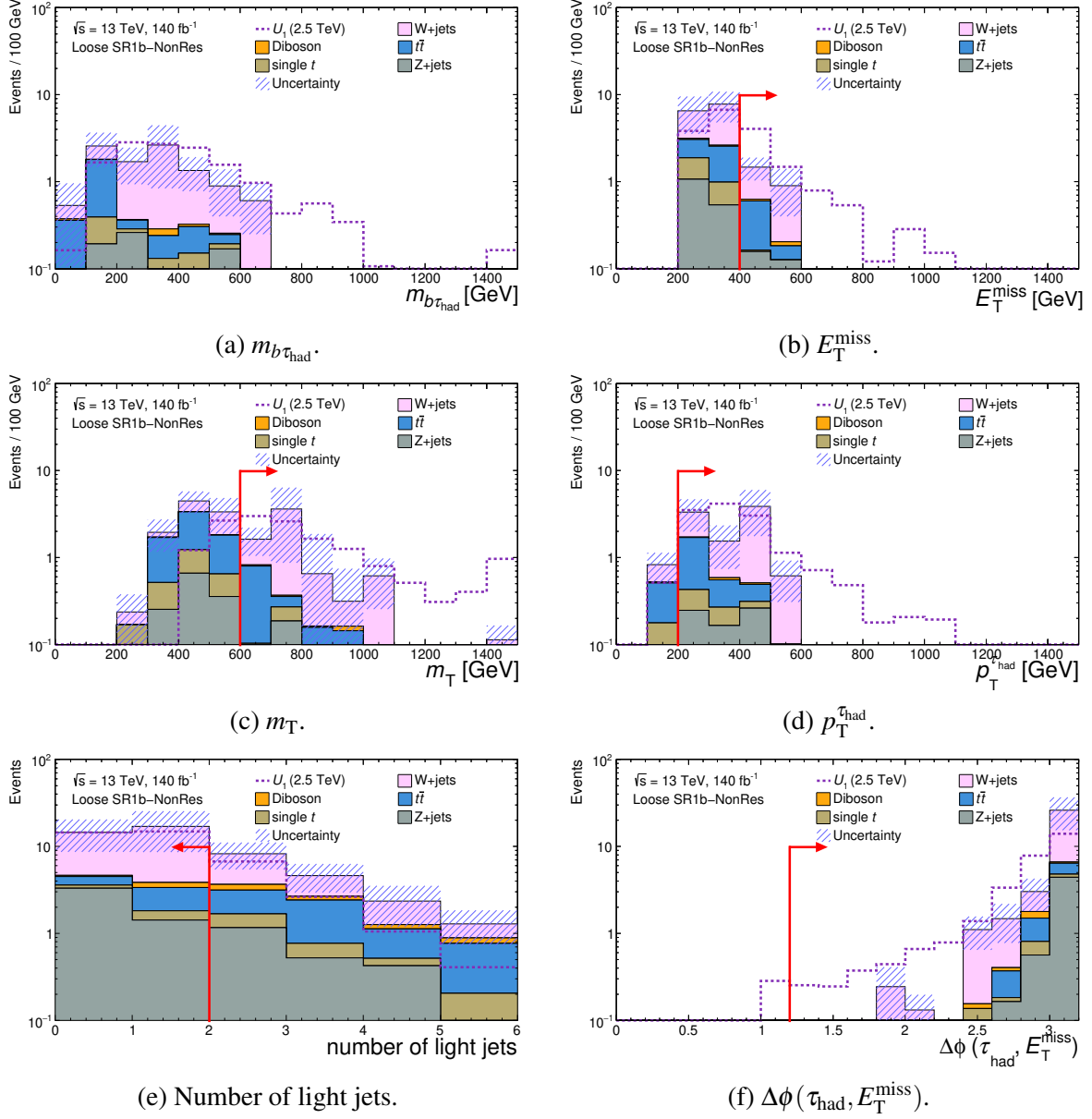


Fig. 7.4 $N-1$ distributions in the loose SR1b-NonRes region. The thresholds are lowered to $m_T > 500$ GeV, $E_T^{\text{miss}} > 300$ GeV and $p_T^{\tau} > 100$ GeV. The baseline $E_T^{\text{miss}} > 200$ GeV is retained. The hatched band represents the background uncertainty, including systematic contributions. The purple dashed line represents the signal at $(M_U, g_U, \beta_L^{23}) = (2.5 \text{ TeV}, 2.5, 1.0)$. The red arrows mark the final selection thresholds and point towards the retained ranges.

7.3 b-veto signal region definition (SR0b)

The SR0b regions is defined based on the SR1b optimisation results, focusing on extracting the on-shell $s\tau$ instead of $b\tau$ LQ decays to extend the sensitivity to large β_L^{23} . The corresponding signal topologies were introduced in Section 3.1.5. In particular, $U_1 \rightarrow s\tau$ produces a resonant signal with a genuine non- b jet, so SR0b retains a physically distinct signal contribution in addition to events in which a b -jet is not tagged or reconstructed. The leading jet in SR0b is not explicitly identified as a strange-quark jet. The truth-level study also identifies a substantial contribution with a leading c jet. At $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$, approximately 30% and 60% of the selected truth-level resonant-category events have a leading s or c jet, respectively. The s -jet contribution contains the $s\tau$ resonance, whereas the c -jet contribution can contain an on-shell $U_1 \rightarrow c\nu_\tau$ decay. For the latter, the resonance is in the $c\nu_\tau$ system rather than the jet- τ system. In the non-resonant truth-level category, approximately 33% and 11% of the events have a radiated leading s or c jet, with most of the remainder containing a gluon jet in $sc \rightarrow \tau\nu_\tau$ production. These fractions describe the chosen benchmark and truth selection, rather than a flavour measurement of the reconstructed jets.

The two SR0b regions are defined by replacing the one- b -jet requirement with $N_b = 0$ and using the leading jet in the topology-dependent selections. The resonant region requires $p_T^{j_1} > 250 \text{ GeV}$ and $m_{j\tau} > 800 \text{ GeV}$, while the non-resonant region requires $50 < p_T^{j_1} < 250 \text{ GeV}$. The p_T^τ , E_T^{miss} , m_T and $\Delta\phi(\tau, E_T^{\text{miss}})$ thresholds follow the corresponding one- b -jet regions. The upper limits are imposed on the total jet multiplicity: $N_j \leq 4$ for SR0b-Res and $N_j \leq 2$ for SR0b-NonRes. These preserve the total jet limits used in SR1b after replacing the tagged jet with an untagged leading jet.

The $E_T^{\text{miss}} > 200 \text{ GeV}$ requirement in SR0b-Res is retained to keep its kinematic selection aligned with SR1b-Res. A supplementary study of tighter E_T^{miss} thresholds is presented in Appendix C. The SR0b-NonRes threshold remains $E_T^{\text{miss}} > 400 \text{ GeV}$, which also maximises the significance in that study.

The destructive interference in the final zero- b -jet selections reduces the nominal pure BSM yield by approximately 20% at $M_U = 1.5 \text{ TeV}$ and by about 30–40% at higher masses. This reduction is appreciable even after the high- m_T and E_T^{miss} requirements, so the final prediction includes the signed interference contribution explicitly. The supplementary E_T^{miss} scan in Appendix C also shows that a higher E_T^{miss} threshold can suppress this contribution.

Figure 7.5 shows the $m_{j\tau}$ distribution before the mass requirement in SR0b-Res. It also shows the transverse-mass distribution above 600 GeV in SR0b-NonRes. The high- $m_{j\tau}$ requirement enhances the resonant signal contribution, while the high- m_T requirement selects non-resonant production. Together with the one- b -jet regions, these selections retain sensitivity to both $b\tau$ and $s\tau$ final states over the coupling range studied in this thesis.

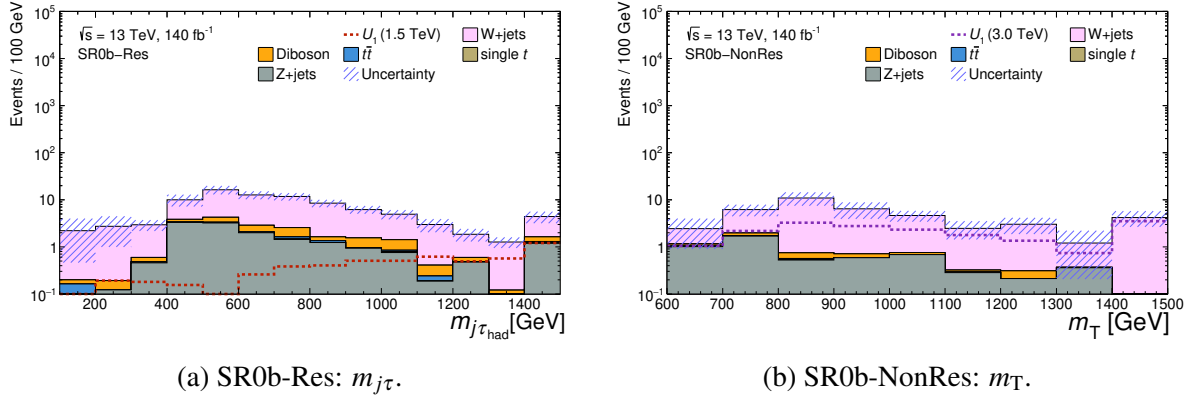


Fig. 7.5 Distributions of (a) $m_{j\tau}$ in SR0b-Res with the mass requirement removed and (b) m_T in SR0b-NonRes above 600 GeV. The other SR requirements are retained. The hatched band represents the background uncertainty. The dotted curves show pure BSM signals at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.5, 0.6)$ in (a) and $(3.0 \text{ TeV}, 2.5, 1.0)$ in (b).

The zero- b -jet background is dominated by $W \rightarrow \tau\nu$ +jets, since vetoing tagged b jets removes much of the top-quark background while retaining W production with light-flavour jets. In the final prediction after the control-region constraints, W +jets contributes approximately 76% in SR0b-Res and 82% in SR0b-NonRes. In the resonant region, diboson production contributes about 10%, $Z \rightarrow \nu\nu$ +jets about 8%, and $Z \rightarrow \tau\tau$ +jets about 5%. In the non-resonant region, the corresponding contributions are approximately 3%, 6% and 8%. The $Z \rightarrow \nu\nu$ events contain a jet misidentified as a τ , while $Z \rightarrow \tau\tau$ and diboson events can contain genuine τ leptons and neutrinos. The top-quark background contribution is below 1% in each zero- b -jet region.

The final requirements for all four mutually exclusive SRs are summarised in Table 7.5. Each region contributes one event count to the simultaneous statistical analysis.

Table 7.5 Common baseline requirements and final signal-region selections. A dash denotes no additional requirement. The jet multiplicity includes b -tagged jets.

Baseline requirements				
Trigger	Lowest-threshold unscaled E_T^{miss} trigger			
N_τ	1			
$N_\ell = N_e + N_\mu$	0			
Multijet suppression	$\min \Delta\phi(E_T^{\text{miss}}, p_T^{\text{jet}}) > 0.4$ or $\frac{E_T^{\text{miss}}}{p_T^{\text{jet}(\min \Delta\phi)}} > 6$			
Requirement	Resonant		Non-resonant	
	SR1b-Res	SR0b-Res	SR1b-NonRes	SR0b-NonRes
N_b	1	0	1	0
p_T^b [GeV]	> 250	–	$50 < p_T^b < 250$	–
$p_T^{j_1}$ [GeV]	–	> 250	–	$50 < p_T^{j_1} < 250$
p_T^τ [GeV]			> 200	
E_T^{miss} [GeV]	> 200			> 400
m_T [GeV]	> 200			> 600
$m_{b\tau}$ [GeV]	> 800	–	–	–
$m_{j\tau}$ [GeV]	–	> 800	–	–
$\Delta\phi(\tau, E_T^{\text{miss}})$	–			> 1.2
N_j	≤ 4			≤ 2

7.4 Tau-pair signal contamination

Leptoquark production with a $\tau\tau$ final state can also populate the $\tau\nu$ + jets signal regions. Although two τ -leptons are produced, an event can satisfy the single- τ requirement when only one is reconstructed and selected. Neutrinos from the τ decays produce missing momentum. This is an additional BSM contribution from the same U_1 model, rather than an SM background.

The study uses simulated $\tau\tau$ samples generated with $\beta_L^{33} = 1$, $\beta_R^{33} = 0$ and $\beta_L^{23} = 0$. Their selected yields are compared with available $\tau\nu$ samples near the exclusion sensitivity, at $(M_U, g_U, \beta_L^{23}) = (1.8 \text{ TeV}, 3.0, 0.6)$ and $(3.0 \text{ TeV}, 2.0, 1.4)$. The corresponding $\tau\tau$ samples use $g_U = 2.8$ and 2.1 . The $\tau\tau$ production cross section is recalculated with MADGRAPH5_AMC@NLO at larger β_L^{23} , and the selected yields are scaled by the cross-section ratio to their generation point. This estimates the additional rate while retaining the acceptance and kinematic shapes of the available samples. It therefore provides a sensitivity study rather than a simulation of the full coupling dependence of the $\tau\tau$ acceptance.

The estimated additional yield is of order 35% of the nominal $\tau\nu$ yield in the one- b -jet regions and several percent in the zero- b -jet regions. To assess the effect on the search, the nominal signal predictions are increased by the factors shown in Table 7.6, and the exclusion fit is repeated.

Table 7.6 Additional signal normalisation used to estimate the sensitivity to $\tau\tau$ production. The factors multiply the nominal $\tau\nu$ signal prediction in the supplementary fit.

Region	Additional contribution	Applied factor
SR1b-Res	37%	1.37
SR1b-NonRes	33%	1.33
SR0b-Res	5%	1.05
SR0b-NonRes	7%	1.07

The expected upper limit on the signal strength improves by approximately 8–14% in this test. The corresponding change in the coupling limit is smaller because the signal rate depends strongly on g_U . At $M_U = 1.8 \text{ TeV}$ and $\beta_L^{23} = 0.6$, the expected upper limit on g_U changes from 2.18 to 2.07, an improvement of approximately 5%. At $M_U = 3.0 \text{ TeV}$ and $\beta_L^{23} = 1.4$, it changes from 1.96 to 1.91, or approximately 2.6%. The nominal interpretation retains the $\tau\nu$ signal model. The supplementary study indicates that the omitted positive $\tau\tau$ contribution could strengthen the expected coupling constraints by a few percent under the assumptions used for the rate estimate.

Appendix B

Multijet background study

The multijet suppression used in Section 7.1.2 is studied with simulated dijet and leptoquark events. The motivation and discriminating variables are introduced there; this appendix presents the study selection, threshold scan and residual-background checks.

B.1 Study selection and distributions

The study selection is kept loose to retain enough simulated multijet events to examine the relevant distributions. It requires the E_T^{miss} trigger, exactly one hadronic τ candidate, no selected electron or muon, and at least one jet, as listed in Table B.1. The final SR thresholds on E_T^{miss} , p_T^τ , m_T and the visible invariant mass are not imposed at this stage. No b -tag multiplicity is required.

Table B.1 Loose selection used to study the multijet suppression. This selection defines the denominator of the rejection fractions shown below.

Requirement	Selection
Trigger	Lowest-threshold unprescaled E_T^{miss} trigger
Hadronic τ multiplicity	$N_\tau = 1$
Electron and muon veto	$N_\ell = 0$
Jet multiplicity	$N_j \geq 1$

Figure B.1 shows the two-dimensional distributions after this selection. The multijet sample is concentrated at small $\min\Delta\phi(j, E_T^{\text{miss}})$ and relatively small $E_T^{\text{miss}}/p_T^{\text{jet}(\min\Delta\phi)}$. The signal occupies a broader angular range and extends to larger momentum ratios. Approximately 90% of the multijet events have $E_T^{\text{miss}}/p_T^{\text{jet}(\min\Delta\phi)} < 6$, compared with approximately 55% of the signal in the study sample. The ratio requirement alone would therefore reject a substantial

fraction of the signal. It is instead combined with the angular requirement so that either a large angle or a large ratio is sufficient to retain an event.

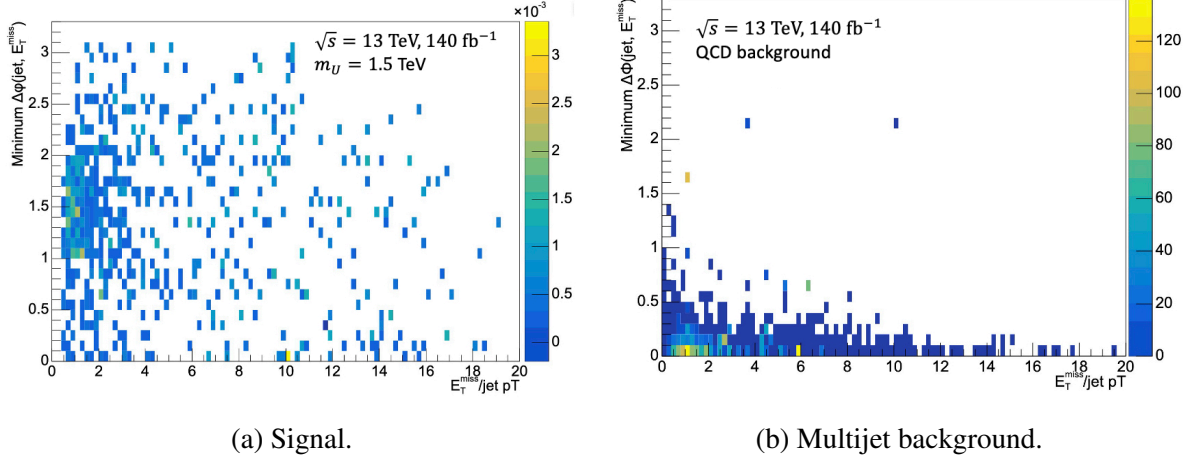


Fig. B.1 Correlation between the minimum jet- $\vec{p}_T^{\text{miss}}$ azimuthal separation and the ratio of E_T^{miss} to the p_T of the closest jet in azimuth, for (a) simulated signal with $M_U = 1.5 \text{ TeV}$ and (b) simulated multijet background after the loose study selection. The concentration of multijet events at small values of both variables motivates a combined requirement.

B.2 Threshold scan and signal retention

The angular threshold is scanned while the momentum-ratio threshold is fixed at six. For an angular threshold X , the rejected fraction is defined as

$$\text{Rejection ratio} = \frac{N\left(\min\Delta\phi(j, E_T^{\text{miss}}) < X \text{ and } E_T^{\text{miss}}/p_T^{\text{jet}(\min\Delta\phi)} < 6\right)}{N(\text{all events after preselection})}. \quad (\text{B.1})$$

The numerator requires both small angular separation and a small momentum ratio. Its complement is the accepted region, defined by the logical OR in Eq. 7.3. The rejected fraction is evaluated separately for signal and multijet events, using each sample's yield after the loose selection as its denominator.

Figure B.2 shows the resulting rejection fractions. At $X = 0.4$, approximately 86% of the multijet events are rejected, while the signal loss is 3.6%. The corresponding signal efficiency for the cleaning cut is therefore $1 - 0.036 = 96.4\%$. Increasing the angular threshold rejects more multijet events but also removes more signal. The selected threshold provides substantial multijet rejection with a small signal loss in the studied sample.

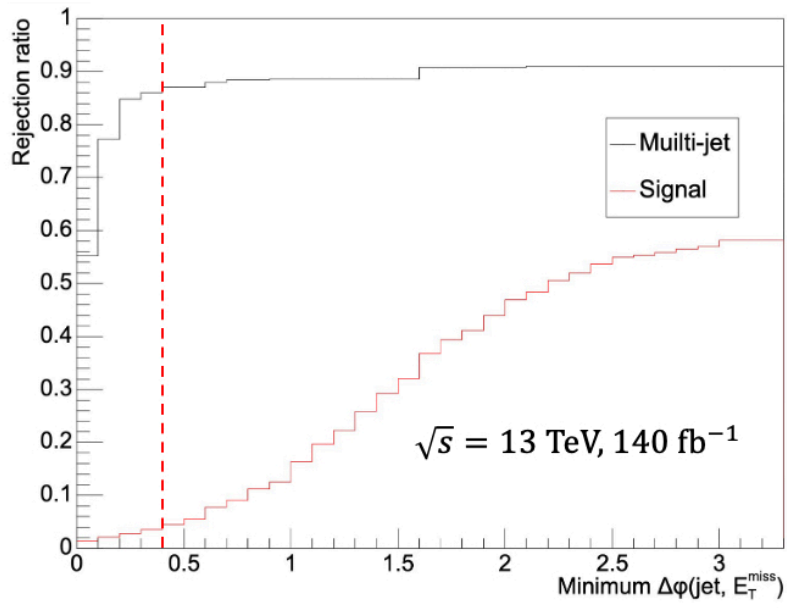


Fig. B.2 Rejected fractions of simulated signal and multijet events as a function of the minimum jet- $\vec{p}_T^{\text{miss}}$ angular threshold X , with $E_T^{\text{miss}}/p_T^{\text{jet}(\min \Delta\phi)} < 6$ also required in the rejected sample. The red and black curves show signal loss and multijet rejection, respectively. The vertical dotted line marks the adopted threshold $X = 0.4$. The denominator is the loose selection in Table B.1.

B.3 Residual multijet contribution

The residual multijet contribution is examined after applying the cleaning cut and the full SR requirements. No simulated multijet events survive those selections in the samples used for the study. This supports neglecting the multijet contribution in the nominal SR prediction, but zero surviving simulated events does not establish an exactly zero physical rate or a quantitative upper bound without the effective sample size.

Jets misidentified as τ candidates can still occur in other background processes. In particular, $Z \rightarrow \nu\nu$ production with jets contains genuine missing momentum and can pass the multijet cleaning even when the selected τ candidate is a jet. This contribution is included in the simulated Z+jets background. A supplementary selection requiring Medium τ identification while failing Tight also gives a negligible multijet contribution in the simulated study, supporting the effectiveness of the cleaning beyond the nominal Tight selection.

Appendix C

Zero-b-jet signal-region optimisation

The E_T^{miss} threshold in SR0b is studied to assess whether tighter selections can improve the sensitivity at large β_L^{23} . A higher threshold can suppress both the SM background and the destructive interference, but also reduces the retained signal. The significance is therefore evaluated as the threshold is varied, including only the leading systematic uncertainties.

The threshold is scanned from 200 to 1500 GeV in 100 GeV steps. The signal parameter points are $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 0.5, 1.8)$ for SR0b-Res and $(3.0 \text{ TeV}, 2.0, 1.8)$ for SR0b-NonRes. The large value of β_L^{23} makes these benchmarks relevant to the SR0b regions discussed in Section 7.3. The scan outcomes and adopted thresholds are summarised in Table C.1.

Table C.1 Signal benchmarks and preferred thresholds in the supplementary SR0b E_T^{miss} scan. The last column gives the thresholds used in the final analysis.

Region	(M_U, g_U, β_L^{23})	Scan maximum	Adopted E_T^{miss} cut
SR0b-Res	(1.5 TeV, 0.5, 1.8)	600 GeV	> 200 GeV
SR0b-NonRes	(3.0 TeV, 2.0, 1.8)	400 GeV	> 400 GeV

For SR0b-NonRes, the significance is maximised at $E_T^{\text{miss}} > 400 \text{ GeV}$. The scan therefore supports retaining the threshold used in the final region definition. For SR0b-Res, the maximum occurs at $E_T^{\text{miss}} > 600 \text{ GeV}$, giving an improvement of approximately 20% in the significance at the chosen benchmark. The final resonant selection nevertheless retains $E_T^{\text{miss}} > 200 \text{ GeV}$ to keep its kinematic requirements aligned with SR1b-Res. The higher threshold also reduces the magnitude of the destructive interference in SR0b-Res. This demonstrates the role of E_T^{miss} in suppressing both the SM background and the interference contribution, although the gain must be balanced against the loss of signal events.

Appendix D

Signal efficiency and interference

Signal efficiencies and the effect of signal–SM interference are examined for the regions defined in Chapter 7.

D.1 Signal efficiency

Signal efficiencies are evaluated using the pure BSM contribution because the signed interference yield is not an event-selection efficiency. The acceptance times selection efficiency, $A\epsilon$, is evaluated with the total generated signal as its reference and includes the kinematic acceptance and reconstruction and selection losses. For illustration, at $(M_U, g_U, \beta_L^{23}) = (1.5 \text{ TeV}, 1.0, 0.2)$ with $\beta_R^{33} = 0$, $A\epsilon$ is 0.30% and 0.13% in SR0b-Res and SR0b-NonRes, compared with 0.64% and 0.069% in the corresponding SR1b regions. These efficiencies depend on the mass and flavour couplings because the relative contributions of the production and decay channels change.

D.2 Contribution to the selected signal yield

The selected signal prediction includes the interference term in Eq. 5.4. If S denotes the pure BSM yield and I the interference yield, the excess above the SM prediction is $S + I$ at the nominal signal normalisation. For destructive interference, $I < 0$, so a selection ranked using S alone can overestimate the final sensitivity. The interference contribution is therefore checked after the optimisation and included in the final statistical model.

D.3 Effect of the kinematic requirements

The low-energy phase space in which interference with SM W production is most important is suppressed by the requirements on p_T^τ , E_T^{miss} and m_T . In particular, the non-resonant requirements $p_T^\tau > 200$ GeV, $E_T^{\text{miss}} > 400$ GeV and $m_T > 600$ GeV retain the hard tail while reducing the destructive contribution.

At $M_U = 1.5$ TeV, the magnitude of the interference contribution in SR1b-Res is typically below 5% of the pure BSM yield over the parameter points studied. In SR1b-NonRes, fractions as large as 40% are found at the lowest g_U values, where the signal yield is too small to be excluded by the analysis. Near the expected exclusion boundary from SR1b-NonRes alone, the contribution is typically below 10% at $M_U = 1.5$ TeV and below 5% at $M_U = 3.0$ TeV. The expected limits obtained with and without interference are found to differ little in the coupling range to which the one- b -jet analysis is sensitive. The pure BSM yield is therefore used to rank the trial selections, while interference is retained in the final signal prediction.

This conclusion does not apply to the SR0b regions, where the interference contribution must be retained, as discussed in Section 7.3. The supplementary E_T^{miss} scan in Appendix C illustrates how a tighter requirement can reduce this contribution in the zero- b -jet regions.

References

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